Quick Start Guide
Low-Noise Wideband Amplifier

LNA-03-30-A-R

1 About This Guide

This document is a condensed quick-start guide intended to support basic installation and operation of the device. This guide does **not** replace the full product manual. The complete operation manual can be downloaded at:
https://www.flimlabs.com/products/low-noise-amplifier/

Scan the QR code to open the full manual.

Reading the complete manual is strongly recommended before operating the device, unless the user is already familiar with similar laboratory instrumentation.

2 Safety Information

Do not operate the device if it is damaged or if safe operating conditions cannot be ensured.

- Read the full manual before installation and use.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.
- Disconnect power before modifying connections.
- Use only undamaged and properly rated cables.

3 Intended Use

The Low-Noise Wideband Amplifier is intended for use in scientific laboratories and research environments as a compact electronic module for amplification of electronic signals. The device is suitable for low-amplitude, high-frequency signals used in scientific instrumentation, photon counting, and fluorescence lifetime imaging microscopy (FLIM) applications.

The device must be operated only by qualified personnel familiar with laboratory electronic instrumentation.

4 Product Overview

The Low-Noise Wideband Amplifier is a compact low-noise broadband amplifier designed for amplification of high-frequency electronic signals. It receives electronic input signals from external devices and provides a non-inverted amplified output signal suitable for downstream acquisition, timing, and analysis instruments.

The device is optimized for signal conditioning in spectroscopy, fluorescence lifetime measurements, and time-correlated photon counting systems.

Key Features

- Low-noise broadband amplification
- One SMA input for the signal to be amplified
- One SMA output for the amplified signal
- Non-inverting amplification
- Broadband operation up to the specified cut-off frequency
- Low timing variation
- USB Type-C power supply
- Compact laboratory form factor

5 Components Overview

Overview of the key components of the device.

1. USB Type-C power input
2. SMA signal input
3. SMA amplified signal output
4. LED power indicator
5. Mounting holes in the enclosure

6 Scope of Delivery

- Amplifier module
- USB Type-C power cable
- Quick Start Guide
- Declaration of conformity

7 Setup

Install the device on a stable laboratory surface. Before installation:

- inspect the device and accessories for visible damage
- verify connectors and cables are intact
- verify that the input signal meets the requirements given in Table 1

8 Connections

Power connection

- Connect the USB Type-C cable to the device power input.
- Connect the other end to a 5 V DC power source.
- Verify that the power LED illuminates.

Input signal connection

- Connect the electronic input signal to the SMA input port.
- Verify that the input signal amplitude does not exceed the specified maximum value.

Output signal connection

- Connect the amplified SMA output to the downstream acquisition, timing, or analysis instrument using a compatible 50 Ω coaxial cable.

9 Configuration

After connecting the device:

- verify that the power LED is illuminated
- verify that the input and output signal connections are secure

The amplifier does not require user adjustment during normal operation. Ensure that the input signal, output connection, and connected downstream electronics are all compatible with the specified signal limits and impedance.

10 Operation

Once the power supply, input signal, and output connections have been correctly established and the initial configuration has been completed, the amplifier operates autonomously without further user intervention. An illustration of signal input and output is show in Figure 1.

11 Further Information

Detailed information on troubleshooting, maintenance, storage, and disposal is provided in the full operation manual available at https://www.flimlabs.com/products/low-noise-amplifier/.

12 Product Label

Low-Noise Wideband Amplifier
P/N: LNA-03-30-A-R
S/N: LNA-XXXX-XXX
5 V ⎓ 0.2 A | IP40

MADE IN ITALY - **FLIM LABS S.r.l.**
Via della Farnesina 3 - 00135 Rome
techsupport@flimlabs.com

13 Technical Specifications

Parameter	Value
Electrical Specifications	
Electrical safety class	Class III
Power supply	USB Type-C, 5 V DC
Current consumption (max)	0.2 A
Power consumption (max)	1 W
Signal Specifications	
Input connections	1x SMA
Input polarity	positive or negative
Maximum input amplitude	5 V
Output connections	1x SMA
Power status indicator	LED
Amplification type	non-inverting
Timing variation	<1 ps
Impedance	50 Ω
Voltage gain	30 dB (>30x)
Cut-off frequency (-3 dB)	2.5 GHz
Low-frequency limit	1 MHz
Physical Specifications	
Dimensions	60 mm × 42 mm × 14 mm
Weight	80 g
Protection rating	IP40
Environmental Requirements	
Operating temperature	0°C to +40°C
Storage temperature	−20°C to +70°C
Humidity	Up to 80% non-condensing

Table 1: Key specifications of the Low-Noise Wideband Amplifier.

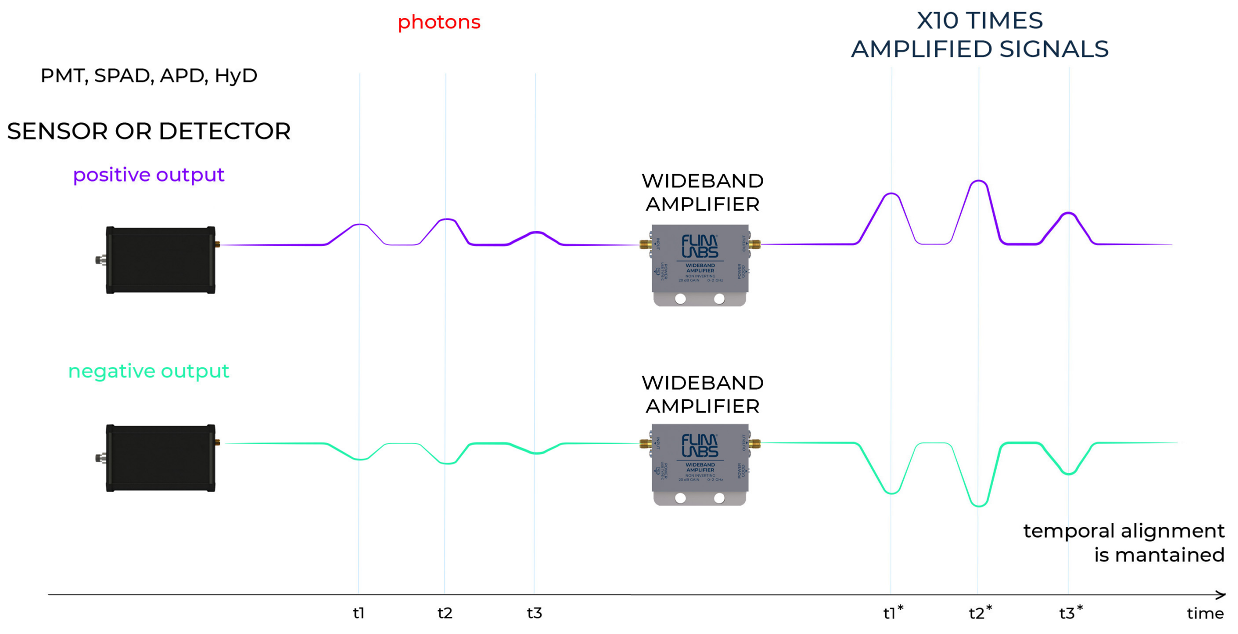

Figure 1: Input and output signal of a configured amplifier